Question: Considering fame of the Pakistani Cinema's New Sensation "The Legends of Muala Jatt "(TLMJ) we are trying to build a sentiment analysis system that will learn from comment section of IMDB. For feature extraction we will be using one hot encoding as discussed in our class. Your Default model consist of 3 layers. Input layer, hidden layer with 2 neurons and a single neuron in output layer with unit cost at each neuron and sigmoid activation function for each neurons.

Your task is to:

1. Draw a RNN Architecture for your given example with respect to time. 5
2. Compute forward pass for the following example. 5

Sample Corpus:

Comments	Label
This movie is EXACTLY needed	1
I can't believe I was able to watch this in theaters	1
Bilal is really talented	1
This bulky look really suits him	1
Hollywood inspired flop movie	0
Mahira khan ruined the show	0
Artificial Punjabi makes it weird	0

Note: Select 8 unique vocabulary words from your relative class's data and transform it to a vector using one hot encoding.

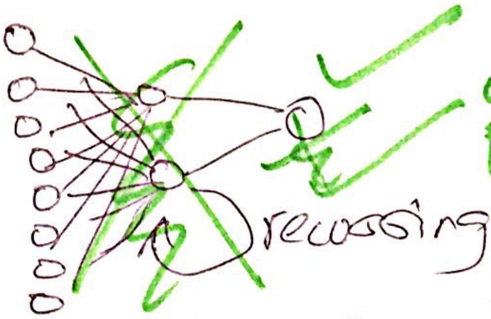
assuming weight and bias as 1

Your Example for Inference:

"Hollywood inspired flop movie"

believe	talented	suits	Hollywood	inspired	flop	movie	[illegible]
1							
	1						
		1					
			1				
				1			
					1		
						1	
							1

believe = $\begin{bmatrix}1\\0\\0\\0\\0\\0\\0\\0\end{bmatrix}$



draw full diagram

$t = 0$

$\sigma(W^T \times I + bias)$

$= \sigma\left(\begin{bmatrix}1&1&1&1&1&1&1&1\\1&1&1&1&1&1&1&1\end{bmatrix}\begin{bmatrix}1\\0\\0\\0\\0\\0\\0\\0\end{bmatrix} + \begin{bmatrix}1\\1\end{bmatrix}\right) = \sigma\left(\begin{bmatrix}1\\10\end{bmatrix}\right)$

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$H^1 = \begin{bmatrix} 0.731 \\ 0.731 \end{bmatrix}$

t=1

talented = $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$\sigma(W^T \times H^2 + W^T \times H^1 + \text{bias})$

$= \sigma\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} 0.731 \\ 0.731 \end{bmatrix} + B\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$

$= \sigma\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1.231 \\ 1.231 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$

$= \sigma\left(\begin{bmatrix} 3.231 \\ 3.231 \end{bmatrix}\right)$

$= \cancel{\begin{bmatrix} 0.962 \\ 0.402 \end{bmatrix}} = \begin{bmatrix} 0.962 \\ 0.962 \end{bmatrix}$

t=2

soits = $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$= \sigma\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} 0.962 \\ 0.962 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$

$= \begin{bmatrix} 0.981 \\ 0.981 \end{bmatrix}$

t=3

Hollywoods $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$= \sigma\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} 0.981 \\ 0.981 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$

$= \begin{bmatrix} 0.981 \\ 0.981 \end{bmatrix}$

t=4

~~soits~~ inspired = $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$= \sigma\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \times \begin{bmatrix} 0.981 \\ 0.981 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$

$= \begin{bmatrix} 0.981 \\ 0.981 \end{bmatrix}$

t=7

$= \sigma\left(\begin{bmatrix} 1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \end{bmatrix} \times \begin{bmatrix} 0.981 \end{bmatrix} + \begin{bmatrix} 1 \end{bmatrix}\right)$

$= \boxed{0.951}$

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